LESSON 8: LINEAR PROGRAMMING

The graphical method: feasible region, corner points, optimal solution

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What is linear programming?

A model of **linear relationships** representing a firm's **objective** and **resource constraints**. It finds the **best (optimal)** way to use **limited resources** to reach a goal by **maximizing or minimizing** a linear **objective function** subject to **constraints**.

THE LP MODEL

- **Objective function** – what you **maximize or minimize**

$$\max / \min \ z = c_1x_1 + c_2x_2 + \cdots + c_nx_n$$

- **Constraints** – inequalities or equations that **restrict** the variables

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n \ (\leq, =, \geq) \ b_i$$

- x_j – **decision variables** | b_i – **constraint levels** (right-hand sides) | c_j – **objective function coefficients** | a_{ij} – **constraint coefficients**

Note: Memory tip: c goes with the **cost/profit** line (objective); **a** sits **among** the constraints; **b** is the **bound** on the right.

TWO APPROACHES

- **Geometric approach – graphical method**: uses the **feasible region** and **corner points** (2 variables)
- **Algebraic approach – simplex method**: developed by **George B. Dantzig (1947)**; for **more decision variables and constraints**

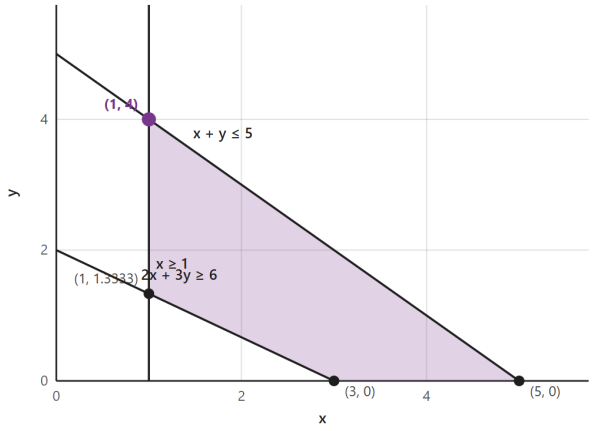
THE GRAPHICAL METHOD

- **1.** Graph each constraint as a **line** (set it to $=$; find the x- and y-intercepts)
- **2.** Shade the side that **satisfies** it (test the point $(0, 0)$: if true, shade toward the origin)
- **3. Feasible region** = where all shadings **overlap** (with $x, y \geq 0$)
- **4. Corner points** = intersections of the boundary lines (solve the pairs of equations)
- **5.** Evaluate the objective function at **every corner**; the **largest** (max) or **smallest** (min) is optimal
- **6. Formulate the decision** in words

Note: The optimal solution is **most likely at a corner point (vertex)** of the feasible region. With only $\geq$ constraints the region can be **unbounded**: a **minimum** exists but **no maximum**.

SLIDE EXAMPLE 1

- Max $P = 30x + 40y$ s.t. $x + y \leq 5, 2x + 3y \geq 6, x \geq 1, x, y \geq 0$
- Corners: $(1, \frac{4}{3}), (3, 0), (5, 0), (1, 4) \rightarrow P = 83.3,\ 90,\ 150,\ 190$
- **Max** $P = 190$ **at** $(1, 4)$



Example 1: feasible region and corner points

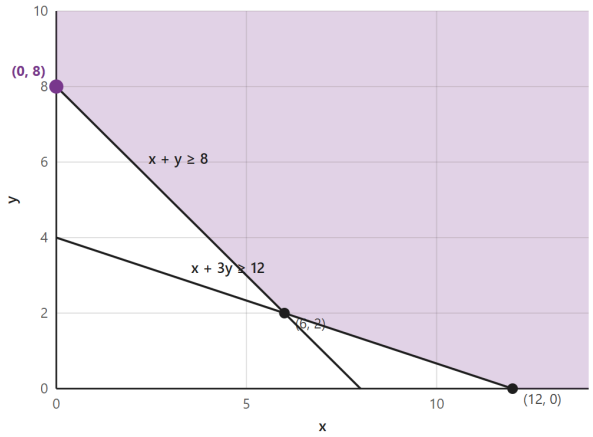
SLIDE EXAMPLE 2

- Max $P = 4x + 5y$ s.t. $x + 2y \leq 10, 6x + 6y \leq 36, x \leq 4, x, y \geq 0$
- Corners: $(0, 0), (4, 0), (4, 2), (2, 4), (0, 5) \rightarrow P = 0,\ 16,\ 26,\ 28,\ 25$
- **Max** $P = 28$ **at** $(2, 4)$

SLIDE EXAMPLE 3 (MINIMIZE)

- Min $C = 5x + 2y$ s.t. $x + 3y \geq 12, x + y \geq 8, x, y \geq 0$
- Corners: $(0, 8), (6, 2), (12, 0) \rightarrow C = 16,\ 34,\ 60$

- **Min** $C = 16$ **at** $(0, 8)$; the region is **unbounded**, so there is **no maximum**



Example 3: an unbounded feasible region

END OF LESSON

LESSON 8 (CONTINUED): FORMULATING LP MODELS

From a word problem to a model and a decision

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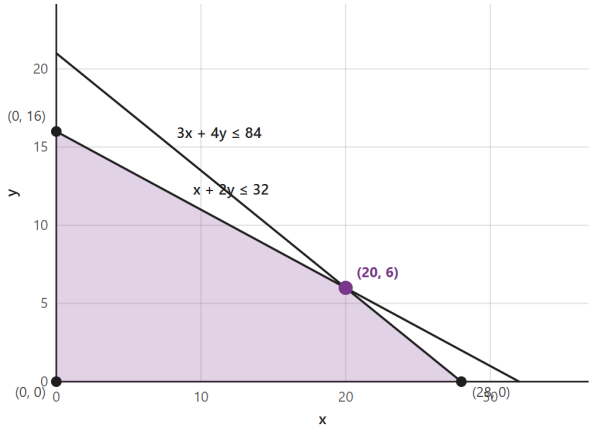
THE 5 STEPS

- **Step 1** – **Read** the problem carefully; organize the data in a **table**
- **Step 2** – **Define the variables** (the unknowns you decide)
- **Step 3** – Write the **objective function** (max or min)
- **Step 4** – Write the **constraint inequalities** (+ **non-negativity**: you can't produce negative amounts)
- **Step 5** – **Solve** (graphically, or with software)

> **Note: Word cues:** "at least / minimum required" → $\geq$ | "at most / maximum available / no more than" → $\leq$ | profit → **maximize** | cost → **minimize**

TENT PRODUCTION (SLIDES)

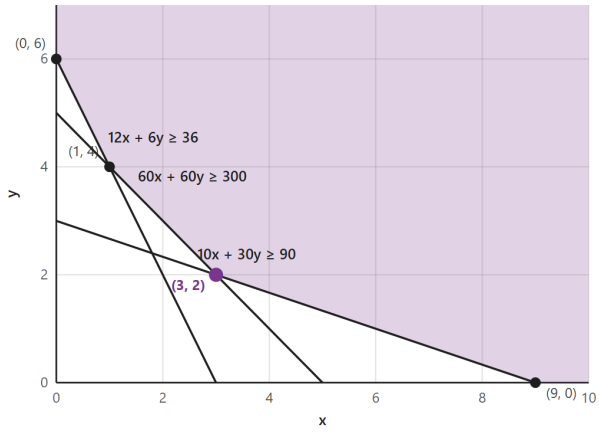
- Standard tent: **1 h** cutting, **3 h** assembly, **₱500** profit | Expedition tent: **2 h** cutting, **4 h** assembly, **₱800** profit | Max hours: cutting **32**, assembly **84**
- Let x = standard tents per day, y = expedition tents per day
- **Max** $P = 500x + 800y$ s.t. $x + 2y \leq 32$ (cutting), $3x + 4y \leq 84$ (assembly), $x, y \geq 0$
- Corners: $A(0,0)$, $B(0,16)$, $C(20,6)$, $D(28,0) \rightarrow P =$ ₱0, ₱12,800, ₱14,800, ₱14,000
- **Decision:** make **20 standard** and **6 expedition** tents daily for a **maximum profit of ₱14,800**



Tent problem: corner C(20, 6) is optimal

DIET DRINKS (SLIDES)

- Drink X: **$0.12**, 60 cal, 12 units vit. A, 10 units vit. C | Drink Y: **$0.15**, 60 cal, 6 units vit. A, 30 units vit. C | Need **at least** 300 cal, 36 vit. A, 90 vit. C
- Let x = cups of X, y = cups of Y: **Min** $C = 0.12x + 0.15y$ s.t. $60x + 60y \geq 300$, $12x + 6y \geq 36$, $10x + 30y \geq 90$, $x, y \geq 0$
- Corners: $(0,6)$, $(1,4)$, $(3,2)$, $(9,0) \rightarrow C =$ $0.90,\ 0.72,\ 0.66,\ 1.08$
- **Decision: 3 cups of X and 2 cups of Y** daily, for a **minimum cost of** $\$0.66$



Diet problem: corner (3, 2) gives the lowest cost

END OF LESSON